YUVRAJ SINGH SIKARWAR

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PROFESSIONAL SUMMARY

B.Tech (ECE) graduate, 2026, specializing in VLSI Design and RTL Coding. Completed a 6-month RTL / Verification internship at Maven Silicon with hands-on work in Verilog, SystemVerilog/UVM, STA, CDC, GLS and DFT, plus RISC-V (RV32I). Built NeuroEdge, a hardware-software co-design pipeline from a quantized CNN to synthesizable Verilog. Seeking an entry-level VLSI / RTL Design Engineer role. Immediate joiner.

WORK EXPERIENCE

RTL / Verification Design Intern | Maven Silicon, Bangalore | Jan 2026 – Jun 2026

- **NeuroEdge Compiler (hardware-software co-design):** built an end-to-end pipeline that trains a CNN on MNIST, applies INT4 / INT8 / mixed-precision quantization (8x / 4x / 7.77x model compression vs FP32) and threshold-optimized pruning (70% sparsity), and generates Verilog for the compute units.
- Designed synthesizable RTL in Verilog: MAC unit (8-bit inputs, 24-bit accumulator), convolution layer, ReLU and top module; verified the MAC with a self-checking testbench and GTKWave waveforms.
- Wrote synthesizable RTL and verified digital logic using SystemVerilog and UVM methodologies.
- Performed Static Timing Analysis (STA), Clock Domain Crossing (CDC) checks and Gate-Level Simulation (GLS) for timing closure.
- Worked on RISC-V (RV32I base ISA) and SoC design flows, applying formal verification and DFT concepts.

KEY PROJECTS

32-bit ALU for RISC-V (RV32I) | Verilog HDL, Icarus Verilog, EDA Playground

- Implemented a 32-bit ALU supporting 8 RV32I operations: ADD, SUB, AND, OR, XOR, SLL, SRL, SLT.
- Designed zero-flag logic to support branch instructions; verified with testbench simulation.
- Structured the code into src/ and testbench/ following industry RTL conventions.

UART Transmitter | Verilog HDL, FSM Design, Icarus Verilog

- Designed an FSM-based UART transmitter with 4 states (IDLE, START, DATA, STOP).
- Implemented 8-bit serial transmission with start/stop framing and a done flag; verified clock generation and reset behavior across multiple bytes.

IoT and Embedded Systems | ESP8266, ESP32-CAM, Arduino

- **Fingerprint Security System:** biometric access control with cloud-connected IoT monitoring.
- **Women's Safety Device:** wearable with GPS tracking and GSM-based SOS alerts, tested across 500m+ range.
- **Face Recognition Door Lock:** real-time face recognition unlock on ESP32-CAM with sub-2-second response.

TECHNICAL SKILLS

RTL and HDL: Verilog HDL, SystemVerilog, RTL Coding, Testbench Development, FSM Design

Verification and Timing: UVM, Formal Verification, Gate-Level Simulation (GLS), Code Coverage, Static Timing Analysis (STA), Clock Domain Crossing (CDC)

Design and SoC: RISC-V (RV32I), SoC Design, VLSI Design Flow, CMOS VLSI, DFT, Low Power Design, Synthesis and Constraints, Physical Design Flow

EDA and Tools: ModelSim, Icarus Verilog, GTKWave, Quartus, Cadence (Basic), EDA Playground, Git, GitHub

Hardware-AI and Embedded: Quantization, Model Pruning, Hardware-Software Co-Design, Arduino, ESP8266, ESP32-CAM

Programming: C, C++, Python, Java

EDUCATION

Manipal University Jaipur | B.Tech in Electronics and Communication Engineering | 2022 – 2026
Relevant coursework: VLSI Design, Embedded Systems, Digital Electronics, Microprocessors, Signal Processing

CERTIFICATIONS

Maven Silicon Advanced VLSI Training Program (17 industry certifications; key modules):

- RISC-V Processor Architecture (RV32I) and System-on-Chip (SoC) Design
- RTL Design using Verilog HDL and Advanced Verilog Coding
- UVM (Universal Verification Methodology) and SystemVerilog for Verification
- Static Timing Analysis (STA), Clock Domain Crossing (CDC) and Gate-Level Simulation (GLS)
- Design for Testability (DFT), Formal Verification and Physical Design Flow